

## Background

- In randomized and blinded clinical trials, the treatment allocated or the treatments corresponding to the arm to which a subject is randomized to is blinded
  - This means, the investigator, study subject and the personnel analyzing the data are not aware of the exact treatment until a certain point in time when the treatments are revealed
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- Once the subject is randomized and ready to be dispensed with medication, investigator uses the IVRS/IWRS to get the 'kit' (medication box) to be dispensed at a particular visit
  - Neither the investigator nor the subject will know what medication is contained in the box (as they are blinded)
  - In the backend, a database will be kept containing the box numbers and what treatment is contained in the box (eg: BOX dataset).
  - BOX dataset will be available for use in programming and during the blinded phase, the contents of each box is populated randomly with the possible treatment values
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- Actual arm variables in SDTM.DM are populated based on the box content received by the subject.
  - Note that sometimes there can be dispensing errors - box lost, damaged box, incorrect box being dispensed etc.
  - There can be different algorithms on how actual arm of a subject is populated.
  - For example, in a multiple dosing study, one example approach is to populate it with most frequently used treatment
  - One other example approach is to use populate it with the first treatment the subject is exposed to

## What do we learn in this lesson?

- In this lesson, we will see how to populate actual arm variables in SDTM.DM domain based on the first treatment the subject is exposed to
- DEMOG dataset - contains a list of all subjects who participated in the study
- IPADMIN dataset - contains the dosing dates and the box number being used at each visit
- BOX dataset - contains the box numbers and associated content in the box

DEMOG

PT

IPADMIN

| PT | Date | IPBOXID |
|----|------|---------|
|    | 1    |         |
|    | 2    |         |
|    | 1    |         |
|    | 2    |         |
|    | 3    |         |

BOX

| KITID | CONTENT |
|-------|---------|
|       | PBO     |
|       | PBO     |
|       | ACTIVE  |
|       | ACTIVE  |
|       | PBO     |
|       | ACTIVE  |
|       | PBO     |

## R programming concepts involved

- The function `as.Date()` is used to **convert a character string into an actual Date object**.
  - `as.Date(ipstdt_raw, format = "%d/%b/%Y")`
  - `%d` stands for two-digit day (e.g., 24)
  - `%b` stands for three-letter month abbreviation (e.g., "SEP")
  - `%Y` stands for four-digit year (e.g., 2023)
- The function `format(dt, format = "%Y-%m-%d")` is used to convert the date value to character notation
  - Here, `format = "%Y-%m-%d"` outputs the date in a standard numeric format like `"2023-09-24"`
- We will also use the concept of replication of first dot processing using `arrange`, `group_by` and `slice` functions